On Solving the Multiple Variable Gapped Longest Common Subsequence Problem

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Abstract. This paper addresses the Variable Gapped Longest Common Subsequence (VGLCS) problem, a generalization of the classical LCS problem involving flexible gap constraints between consecutive solutions' characters. The problem arises in molecular sequence comparison, where structural distance constraints between residues must be respected, and time-series analysis, in which events are required to occur within specified temporal delays. We propose a novel search framework based on a root-based state graph representation, in which the overall state space comprises of generally large number of rooted state subgraphs. To cope with the resulting combinatorial explosion, an iterative beam search strategy is employed, dynamically maintaining a global pool of promising candidate root nodes, enabling effective control of diversification across iterations. To explore the search, several heuristic evaluations inspired by the LCS literature are incorporated into the standalone beam search procedure. To the best of our knowledge, this work presents the first comprehensive computational study of the VGLCS problem, generating a benchmark of 320 synthetic instances with up to 10 input sequences and up to 500 characters. Experimental results show robustness of the novel approach over the baseline beam search in a comparable runtime.

Keywords: Longest common subsequences · Beam Search · Bioinformatics · Gap constraints

1 Introduction

The *Longest Common Subsequence Problem* (LCSP) [1, 4] is a well-known combinatorial optimization problem with numerous applications in bioinformatics

and computational biology, playing a fundamental role in the analysis and comparison of molecular sequences. Given an arbitrarily large set of input sequences $S = \{s_1, \ldots, s_m\}, m \in \mathbb{N}$ over an alphabet Σ, the objective is to identify a longest possible subsequence that is common to all sequences $s_i \in S$. Over the past decades, a variety of practically motivated extensions of the LCSP have been proposed to better capture structural, biological, or application-specific requirements of real-world sequences. Notable variants include constrained, arc-preserving, and repetition-free longest common subsequence problems [6, 5, 3], among others. In this work, we focus on yet another practical variant, the *Variable Gapped LCS Problem* (VGLCSP), originally introduced for the $m = 2$ case in [9], later investigated from a theoretical perspective in [2]. The VGLCSP extends the classical LCSP by incorporating flexible distance constraints (referred to as *gaps*) between consecutive symbols of the resulting subsequence. Unlike fixed-gap models, these gap limits are allowed to vary along the input sequences, thereby offering increased modeling flexibility.

Formally, for each input sequence $s_i \in S$, a gap constraint is defined by a function at positions of sequence as $G_i \colon \{1, \ldots, |s_i|\} \mapsto \mathbb{N}$. Precisely, if the characters of solution s occur at positions $i_i^1 < \cdots < i_i^{|s|}$ in s_i, the gap constraint is fulfilled iff $i_i^x - i_i^{x-1} \leq G_i[i_i^x] + 1, \quad x = 2, \ldots, |s|$. A common subsequence s is considered *feasible* if the corresponding gap constraints are satisfied simultaneously for all sequences in S. As an example of a VGLCSP, given is a set of two input sequences $S = \{s_1 = \texttt{ABCA}, s_2 = \texttt{ACAB}\}$ with two gap constraints $G_1(\{1, \ldots, 4\}) \mapsto \mathbf{1} = G_2(\{1, \ldots, 4\})$. The sequence $s = \texttt{ACA}$ is a feasible solution because it is a common subsequence of the both input strings, where its corresponding characters appear at positions 1, 3 and 4 in s_1 and at the positions 1, 2, and 3 in the s_2. As the difference between each consecutive pair of the indices of these positions is less or equal $1 + 1 = 2$, s represents a feasible sequence of the tackled problem. On the other hand, sequence $s =$AB is also a common subsequence, but not fulfilling all gap constraints. In details, the corresponding characters appear at positions 1 and 2 in s_1 and at positions 1 and 4 in s_2. Obviously, $4 - 1 = 3 > 2 = 1 + 1$ thus violating the second gap constraint. The VGLCSP provides a flexible and realistic model for sequence comparison, particularly suited to applications in DNA and protein analysis where variable structural distances between (active) residues must be respected. Beyond bioinformatics, the problem also arises in time-series analysis, especially in settings where events are required to occur within specified temporal delays [7].

Although several dynamic programming approaches exist for the $m = 2$ case [9], they are computationally prohibitive when expanded for larger m. To the best of our knowledge, no effective exact or approximate approaches are known for the generalized VGLCSP with arbitrary m, clearly an $\mathcal{NP}$-hard problem as a generalization of the LCS problem with arbitrary large set of input strings.

Preliminaries. Let $|s|$ denote the length of a sequence s, and let $s[i]$ refers to the character of s at position i, where indexing starts at $i = 1$. The substring of s that begins at position i and ends at position j is denoted by $s[i, j]$. If $i = j$, this corresponds to the single character $s[i]$; if $j < i$, then $s[i, j] = \varepsilon$, where

ε denotes the empty string. The number of occurrences of $a \in \Sigma$ in s is denoted by $|s|_a$. We denote by $S = \{s_1, \ldots, s_m\}$ the set of input sequences, and by $S^{rev} = \{s_1^{rev}, \ldots, s_m^{rev}\}$. Unless stated otherwise, $m \in \mathbb{N}$ is the number of input sequences (and, correspondingly, the number of gap constraints), and n denotes the length of the longest sequence in S. Given is a positional vector $\mathbf{p}^L \in \mathbb{N}^m$, subproblem related to these positions is given by $S[\mathbf{p}^L] = \{s_i[\mathbf{p}_i^L] \mid i = 1, \ldots, m\}$. Last but not least, for a gap constraint G_i, its reverse constraint is denoted by G_i^{rev}, and is given as $G_i^{rev}(j) := G_i(|s_i| - j + 1)$, for each $j = 1, \ldots, |s_i|$.

The remainder of the paper is organized as follows. Section 2 introduces the rooted state graph representation of the problem. In Section 3, we design of our main methodological contribution, the iterative multi-source beam search approach. Section 4 reports the computational results, comparing the proposed method with a baseline beam search and an iterative greedy heuristicFinally, Section 5 concludes the paper and outlines directions for future research.

2 Rooted State Space Formulation

Building on the state-space representation for the LCSP introduced in [1], we develop a *rooted state graph* model for the VGLCSP. Each state represents one or more feasible partial solutions characterized by a vector of positions induced by the input sequences and by the current subsequence length.

Formally, a partial subsequence s^v induces a state node $v = (\mathbf{p}^{L,v}, l^v)$, where $\mathbf{p}^{L,v} = (p_1^{L,v}, \ldots, p_m^{L,v})$ and the following conditions hold: (i) for each sequence $s_i \in S$, the index $p_i^{L,v} - 1$ is the smallest position such that s^v is a subsequence of $s_i[1, p_i^{L,v} - 1]$; (ii) $l^v = |s^v|$ denotes the length of the partial subsequence; and (iii) all gap constraints G_i are *satisfied* by s^v with respect to each input sequence s_i. A direct arc $\alpha = (v_1, v_2)$, labeled with $\texttt{let}(\alpha) = a \in \Sigma$, exists if $l^{v_2} = l^{v_1} + 1$ and $s^{v_2} = s^{v_1} \cdot a$, that is, state v_2 corresponds to extending a partial solution of v_1 by appending the character a, respecting the gap constraints.

To expand a state v, those characters $a \in \Sigma$ that occur in the suffix of each sequence s_i starting at position $p_i^{L,v}$, for $i = 1, \ldots, m$, are first detected. For each such $a \in \Sigma$, we identify the smallest feasible positions $p_i^{L,a} \geq p_i^{L,v}$ such that $s_i[p_i^{L,a}] = a$ and $p_i^{L,a} - p_i^{L,v} \leq G_i(p_i^{L,a}), i = 1, \ldots, m$. In that way, a child state $w = (\mathbf{p}^{L,a} + \mathbf{1}, l^v + 1)$ is created, unless it is dominated by another child state according to the dominance criteria defined later in this section. To efficiently compute children and their associated position vectors, we employ a preprocessing data structure denoted by `Succ`: a three-dimensional integer array indexed by (i, j, a), where i refers to the input sequence s_i, $1 \leq j \leq |s_i|$ denotes a position within that sequence, and $a \in \Sigma$ is a character. As already pointed out, the value $q = \texttt{Succ}[i, j, a]$ stores the smallest index $q \geq j$ such that $s_i[q] = a$ so that the gap constraint is satisfied at position q, that is, $q - j \leq G_i(q)$. If no such position exists, the value -1 is assigned. Goal states are those that cannot be further expanded. The trivial root state is given by $r = ((1, \ldots, 1), 0)$ as in-going edges to this node are not possible, representing an empty partial

solution as in the case of LCS problem. However, it is not the only root node, in general. The problem can have many (possibly exponentially) root nodes due to the definition of the problem itself allowing a freedom to choose leading letter of partial solutions, induced from the position $u = (i_1^1, \ldots, i_m^1)$ matching that letter, as gap constraints do not apply to the starting letter of each solution.

This motivate us to introduce the state space subgraph rooted with the match $p^{L,u}$, denoted by $\textsc{RootedSubgraph}(u = (p^{L,v}, 0))$, induced by empty partial solution, which consists of all nodes (and corresponding transitions) that can be feasibly reached via node $(p^{L,v} + \mathbf{1}, 1)$, constructed directly from the root node u through its extension by the letter at the position $p^{L,u}$. In terms of feasible partial solutions, $\textsc{RootedSubgraph}(u)$ will generate solutions whose leading letter is the letter at the match $p^{L,u}$, recursively repeating the procedure of expansion of solutions by one letter in all possible ways, until no more expansions are possible. As discussed in the following section, the structure of this subspace heavily depends on the choice of the root state from which the search is initiated. As an example, we illustrate the state space of $\textsc{RootedSubgraph}(u = (1,1), 0)$, depicted in Fig. 1. Note that here the position $(1,1)$ correspond to the match of letter `A`, where $u = ((1,1), 0)$ indeed represents a root node.

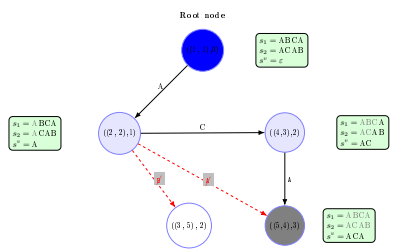


Fig. 1: State space $\textsc{RootedSubspace}(u = ((1,1), 0))$ between the sequences $\boxed{\texttt{A}}$`BCA` and $\boxed{\texttt{A}}$`CAB`, assuming $G_1 = G_2 = \mathbf{1}$. A final/terminal state is displayed in gray. The white-filled node is a valid LCS node, but not an VGLCS node, as the second gap constraint has been violated. The longest solution generated from this subspace is `ACA`.

Multi-Source Beam Search. Unlike the classical LCSP, the VGLCSP may exhibit multiple, potentially exponentially many, *disconnected root components* in its complete state space. Consequently, a search strategy that expands states from a single initial root may fail to reach the optimal solution.

As an illustrative example given in Fig. 2, consider the sequence set $S = \{s_1 = \texttt{ATGGAAA},\ s_2 = \texttt{ATCCAAA}\}$, with $G_{s_1} = G_{s_2} = 1$. In this instance, any state with position vector $\mathbf{p}^L = (5,5)$ cannot be reached from the initial state $((1,1), 0)$ by a sequence of feasible direct transitions. As a result, the optimal common subsequence `AAA` is unreachable from the root $((1,1), 0)$ when exploring its subgraph. To address this issue, the complete state space of an VGLCSP instance is defined by $\bigcup_{v \in \mathcal{R}} \textsc{RootedSubspace}((\mathbf{q}^{L,v}, 0))$ where $\mathcal{R}$ denotes the set of all root states, which cannot be reached from any other state by a series of direct transitions. However, explicitly enumerating all such root states is compu-

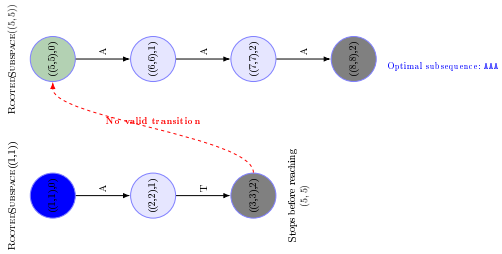


Fig. 2: Root state space subgraphs: disconnected components.

tationally prohibitive in general, requiring $O(n^m)$ time, and thus infeasible for instances with large m or n. This structural characteristic of the VGLCSP motivates the methodology entitled *Iterative Multi-Source Beam Search* (IMSBS) approach, specifically designed to dynamically explore multiple promising rooted subregions of the state space by systematically moving from one rooted subgraph to another one. By iteratively identifying and expanding a set of candidate root states, IMSBS effectively mitigates the disconnection effects induced by gap constraints, enabling the discovery of high-quality solutions that would otherwise remain inaccessible to single-source search methods.

3 Iterative Multi-source Beam Search Framework

Beam search (BS) is a well-established tree-search-based metaheuristic that operates in a breadth-first-search manner. The search is controlled by a beam width parameter β, which specifies the maximum number of nodes retained for expansion at each levels, and by a heuristic guidance h, which guides the selection of most promising candidates for further expansion. Initially, the procedure receives a set of nodes as the initial beam, which are then expanded, generating a set of candidate nodes for a beam of the subsequence level. After being evaluated, up to β most promising nodes are kept for in the beam of the subsequent level. The procedure repeats until beam is empty, returning the longest partial solution.

In the IMSBS, beam search component is served to intensify the search process. As the search space of an VGLCSP instance is represented by the set of root nodes $\mathcal{R}$, BS procedure is in charge for efficiently exploring the search subspace $\textsc{RootedSubspace}((\mathbf{p}^L, 0))$ induced by the corresponding root node. Note that one could here employ an exact technique (such as A* or BFS); however, solving the subproblem itself could be a challenge task, as hard as the original problem itself. Thus, we have opted for more robust BS technique.

The overall IMSBS framework is summarized as follows. Initially, the global pool of candidate root nodes is initialized to an empty set $\mathcal{R} := \{\}$ and the current best solution to $s_{\text{best}} := \varepsilon$. First, for each $a \in \Sigma$ the closest matches $\mathbf{p}^{L,a}$ from the leading positions of each input sequences are detected. Each such (non-dominated) match generates one root node $(\mathbf{p}^{L,a}, 0)$, which has been subsequently added to $\mathcal{R}$. The core iterations of the IMSBS are then executed until

either $\mathcal{R}$ becomes empty, a predefined maximum number of iterations is reached, or allowed time limit is exceeded. At each IMSBS iteration, a fixed number ($sources_from_R$) of most promising candidate root nodes according to a heuristic criterion h' are taken from $\mathcal{R}$, subsequently stored in a temporary beam $\mathcal{L}$. These nodes are further refined, executing a single backward beam search procedure for each $w \in \mathcal{L}$ with a root node $(\mathbf{p}^w, 0)$. The procedure works similarly as the baseline BS, expanding nodes with accordance to the gap constraints but in the reversed manner, starting from the root positions at each sequence and moving the search (pointers) towards their leading positions. In essence, for each $w \in \mathcal{L}$, the same RootedBeamSearch is executed on the reversed input sequences S^{rev} and the reversed gap constraints $(G_i^{rev})_{i=1}^m$ with the initial beam width $B = \{((|s_i| - p^w)_{i=1}^m, 0)\}$. Assume the procedure has completed, returning the solution s_{bwrd}^w. In this way, the reverse sequence s_{bwrd}^{rev} is a feasible VGLCS solution. The partial solution without the last letter—which correspond to the match $\mathbf{p}^w$—is assigned to the refined node $w \in \mathcal{L}$, while the updated $w = (p^w, |(s_{bwrd}^w)^{rev}| - 1)$ is passed to the forward beam search procedure to another round of extensions. Thus, the refined set $\mathcal{L}$ enters in as the initial beam, executing a BS parametrized with β and $h(v)$, following the subspace defined in Section 2. During this phase, all completed (non-expandable) nodes are collected in a list $V_{complete}$, being returned along with the found solution s_b after its completion, verified for a new incumbent. Each complete node from $V_{complete}$ is then independently expanded—ignoring the gap constraints— to generate a set of candidate root nodes. In details, for each complete node $v = (\mathbf{p}^v, l^v)$, candidate root nodes $r^{w,a} = (\mathbf{p}^{w,a}, 0)$ are generated for each letter $a \in \Sigma$ consisting of positions $p_i^{w,a}$ that matches letter a in s_i which are the closest to the position $\mathbf{p}_i^v$. Each such candidate node $(r^{w,a}, 0)$ is added in the pool $\mathcal{R}$, provided that it has not already been part of $\mathcal{R}$ during the previous iterations. Once the BS procedure is terminated, a new Imsbs iteration is initiated with a newly selected set of most promising root nodes from $\mathcal{R}$, until a termination condition has met. The pool $\mathcal{R}$ may be maintained as a priority queue implemented via a binary heap, to ensure efficient extraction of the most promising candidate root nodes. The core advantages of the IMSBS approach over the baseline BS are: (i) A provision of balance between complete beam search execution (local exploration of promising subgraphs) and the generation of new candidate root nodes (global coverage of the search space); (ii) Prevention from suboptimal candidate root nodes reduced by executing backward beam search procedure.

Heuristic guidances. Several heuristic guidances to guide search are utilized within the IMSBS—as the candidates to guide froward and backward BS, h and h' respectively. These are known from the LCS literature: (i) *"Look-ahead" for the remaining length:* $\mathrm{UB}_1(\mathbf{v}) = l^v + \min_{i=1,\ldots,m}\left(|s_i| - p_i^{L,v} + 1\right)$; (ii) *Character Frequency Alignment:* $\mathrm{UB}_2(\mathbf{v}) = l^v + \sum_{\sigma\in\Sigma} \min\left(|s_1[p_1^{L,v}, |s_1|]|_\sigma, \ldots, |s_m[p_m^{L,v}, |s_m|]|_\sigma\right)$, counting the maximum number of characters that can still match in the (optimal) solution based on the frequency of each character; (iii) *A probability-based heuristic* guidance h_{prob} by the Mousavi and Tabataba's matrices [8]: These

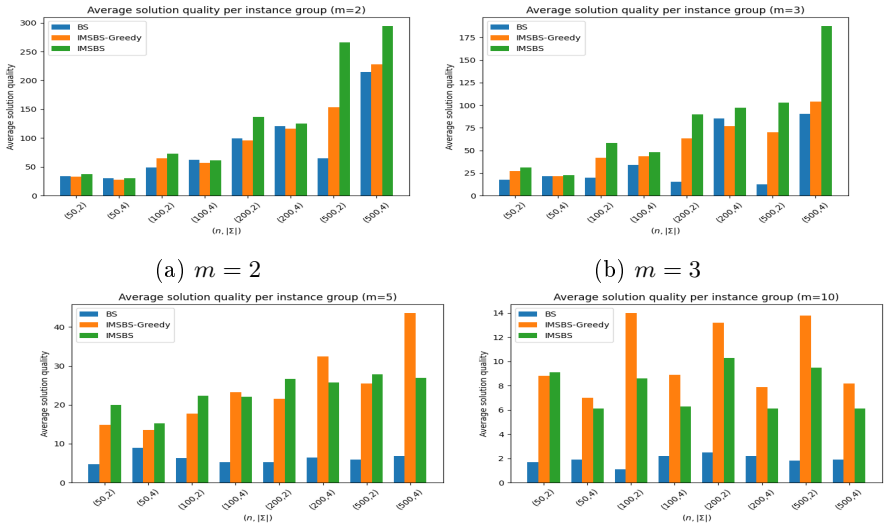
probabilities are based on the idea of determining a probability of the event that a sequence s of length k is a subsequence of a (random) sequence of length n over Σ. More about these heuristics the interested readers can found at [1, 8].

4 Experimental Evaluation

We compare the performance of heuristic approaches designed for an arbitrary $m \geq 2$. Specifically, the following methods are compared: (i) Bs: a baseline beam search, which performs a single iteration of IMSBS while employing a huge beam width; (ii) Imsbs-greedy: it sets a fixed $\beta = 1$ employing a huge number of IMSBS iterations; (iii) Imsbs: a tuned IMSBS with an average runtime comparable to that of Bs. The Imsbs is implemented in Python 3.11 and evaluated on the VEGA HPC system hosted at IZUM, Maribor. The cluster consists of 960 compute nodes equipped with AMD EPYC 7H12 CPUs operating at 2.35 GHz. All experiments were conducted in a single-threaded setting. For each combination of $n \in \{50, 100, 200, 500\}$, $m \in \{2, 3, 5, 10\}$, and $|\Sigma| \in \{2, 4\}$, 10 random problem instances are generated. First, m sequences of equal length are generated uniformly at random. Then, the gap constraints for each instance are generated such that each value $G_i(j), j = 1, \ldots, |s_i|, i = 1, \ldots, m$ is assigned uniformly from $\mathcal{U}(\{\lfloor 0.5 \cdot |\Sigma| \rfloor, \ldots, \lfloor 1.5 \cdot |\Sigma| \rfloor\})$. Therefore, a total of *320 problem instances* is generated. The benchmark set is denoted as Random, freely available at `https://github.com/markodjukanovic90/NikolaBalabanDiplomski/tree/main/src/instance_i_generator`.

Parameter tuning. To tune the parameters of our algorithms, we employed a basic grid search to more sensible parameters: beam width in the beam search forward pass (β), and heuristic guidance $h \in \{\mathrm{UB}_1, \mathrm{UB}_2, h_{prob}\}$, comparing the average solution quality over all (320) problem instances. Less sensible parameters are chosen after several preliminary runs. We fix $sources_from_R = 10$ and $beam_iters = 100$, and $h' = \mathrm{UB}_2$. For the baseline Bs approach, a high beam width $\beta = 10,000$ and $h = h_{\mathrm{prob}}$ are utilized. Remaining parameters of the IMSBS approach received the following values: $h = \mathrm{UB}_2$ and $\beta = 500$ to keep comparable average runtime between IMSBS and the baseline Bs approach. These values balance runtime relative to Bs while maintaining reasonable solution quality. Similarly, for the Imsbs-greedy approach with fixed $\beta = 1$ uses $beam_iters = 10000$, and all other parameters hold as in the tuned Imsbs.

Numerical results for all three heuristic approaches are displayed in Fig. 3, reporting the average solution quality per instance group for each approach. Four plots are provided, one per each value of m. The Imsbs approach produces the highest average solution quality in 21 out of 32 instance groups, surpassing the baseline Bs approach dramatically, indicating the selection of a right root node is essential in achieving high-quality solutions. Bs performs better in just one case. Concerning the Imsbs-greedy, it surpasses the other two approaches in 10 (out of 32) cases, mostly for the instances with larger $m = 10$, having shorter final solutions where it turns out that frequently moving from one to another root and exploring respective subgraphs of the search space is strategically cru-



(a) $m = 2$ (b) $m = 3$

(c) $m = 5$ (d) $m = 10$

Fig. 3: Average solution quality of the three approaches.

cial than just strengthening intensification around a solution with a high beam width. Exploring additional root nodes improves the diversity of the search process, necessary in the case when expected solutions are short. This is exactly where IMSBS-GREEDY fits well. BS and IMSBS approaches are tuned to exhibit comparable average runtimes. IMSBS-GREEDY has a significantly larger average runtime due to a higher number of executed iterations ($beam_iters = 10,000$) for the instances with larger m, in contrast to by iterations less intensive approaches. The detailed numerical results with accompanying plots can be found in the supplementary material at `XX`.

5 Conclusions and Future Work

This paper studied a generalized variable gapped LCS problem. It extends the classical LCS problem by constraining the allowable distances between matched characters, reflecting biological scenarios in which spatially closer nucleotides interact more strongly than distant ones. We proposed an Iterative Multi-Source Beam Search (IMSBS) which performs local exploration from promising root nodes involving a global strategy of candidate nodes selection based on previously collected search information. Experimental results indicate the IMSBS consistently produces highest-quality solutions over the baseline beam search. For the instances with a shorter expected solutions, the strategy of frequently moving from one feasible subspace to another one appears more efficient, than in case of small sequence number, where a high beam width and number of IMSBS iterations is required in achieving high-quality solutions. Future research includes designing more sophisticated heuristics which incorporate gap constraints into scoring, optimizing IMSBS to efficiently reuse previously explored regions of the state space, and testing the approaches on real-world biological instances.

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